

FUNCTIONAL REQUIREMENTS DOCUMENT

Restaurant Management System (RMS)

Detailed functional specification derived from the approved Business Requirements Document (BRD v1.0)

DOCUMENT	Functional Requirements Document (FRD)	VERSION	1.0 (Draft for review)
DATE	7 July 2026	CLASSIFICATION	Confidential
PREPARED BY	Project Team	TRACES TO	BRD v1.0 (RMS)
PHASE	Phase 1 (Desktop) → Phase 2 (Web)	STATUS	Draft for review

Document Control

Revision History

Version	Date	Author	Description of change
0.1	5 Jul 2026	Project Team	Initial skeleton and section outline.
0.9	6 Jul 2026	Project Team	Full functional specifications drafted for all six modules.
1.0	7 Jul 2026	Project Team	Complete draft released for stakeholder review; traceability matrix added.

Reviewers & Approvers

Role	Name	Responsibility	Signature / Date
Project Sponsor / Owner		Approves scope and business fit	
Restaurant Manager (Business Owner of requirements)		Confirms functional correctness	
Development Lead		Confirms technical feasibility	
QA / Test Lead		Confirms testability of acceptance criteria	

Related Documents

Ref	Document	Relationship
REF-1	Business Requirements Document (BRD) — RMS v1.0	Parent document; source of business objectives and FR-01...FR-30.
REF-2	Development Plan (companion)	Captures technology and design decisions (JavaFX, MySQL, MVC).
REF-3	Conceptual & Logical Database Design	Detailed entity attributes and 3NF schema (design phase).
REF-4	Test Plan & Traceability Matrix	Maps each FR to test cases for verification.

Table of Contents

1. Introduction	4
1.1 Purpose of this Document	4
1.2 Scope of the System	4
1.3 Intended Audience	4
1.4 Definitions, Acronyms & Abbreviations	4
1.5 Document Conventions	5
2. System Overview	6
2.1 System Context	6
2.2 Architecture & Phasing	6
2.3 Actors	6
2.4 Role–Permission Matrix	6
3. Functional Requirements Specification	8
3.1 Authentication & Administration (FR-01...FR-04)	9
3.2 Menu & Table Management (FR-05...FR-09)	13
3.3 Orders & Billing / POS (FR-10...FR-17)	17
3.4 Inventory & Suppliers (FR-18...FR-22)	23
3.5 Staff & Reservations (FR-23...FR-26)	27
3.6 Reporting (FR-27...FR-30)	30
4. Key Use Cases & Process Flows	33
5. Data Requirements & State Models	35
6. Business Rules Catalogue	37
7. Non-Functional Requirements	39
8. Assumptions, Dependencies & Constraints	40
9. Requirements Traceability Matrix	41
10. Acceptance & Success Criteria	43
Appendix A — Field Validation Reference	44
Appendix B — Status & Enumeration Reference	45

1. Introduction

1.1 Purpose of this Document

This Functional Requirements Document (FRD) translates the business needs expressed in the approved Business Requirements Document (BRD v1.0) into precise, testable functional specifications for the Restaurant Management System (RMS). Where the BRD answers *what* the business needs and *why*, this FRD answers *how the system must behave* to satisfy those needs — the inputs it accepts, the rules it applies, the outputs it produces, the validations it enforces, and the conditions under which each function succeeds or fails.

Every functional requirement identifier introduced in the BRD (FR-01 through FR-30) is carried forward unchanged and expanded here into a full specification. This preserves end-to-end traceability from business objective → functional requirement → use case → acceptance criterion → test case. This document deliberately avoids prescribing internal implementation detail (class design, SQL, screen pixel layout); those belong to the Development Plan and Database Design (REF-2, REF-3).

1.2 Scope of the System

The RMS automates the core day-to-day operations of a single-location restaurant across six functional modules: **Authentication & Administration**, **Menu & Table Management**, **Orders & Billing (POS)**, **Inventory & Suppliers**, **Staff & Reservations**, and **Reporting**. It is delivered in two phases against one shared MySQL database: Phase 1 is an on-premise desktop application (JavaFX), and Phase 2 re-uses the same business logic and data layer behind a web interface.

Out of scope for the current phases (and therefore not specified here): online customer-facing ordering and delivery, third-party payment gateway / card processing, kitchen display screens, loyalty / rewards programs, accounting or payroll integration, multi-branch consolidation, and a native mobile app. These may be considered as future phases.

1.3 Intended Audience

Audience	How they use this document
Development Team	Primary specification for building each function correctly.
QA / Test Lead	Source of testable acceptance criteria and validation rules.
Restaurant Manager / Owner	Confirmation that the system will support real operational workflows.
Project Sponsor	Verification that scope aligns with agreed business objectives.

1.4 Definitions, Acronyms & Abbreviations

Term	Meaning
RMS	Restaurant Management System — the software specified here.
POS	Point of Sale — the ordering and billing module.
BRD / FRD	Business / Functional Requirements Document.
BO-n	Business Objective identifier from the BRD (BO-1...BO-6).
FR-nn / NFR-nn	Functional / Non-Functional Requirement identifier.

BR-nn	Business Rule identifier defined in §6 of this FRD.
Order line / Order Item	A single menu item plus quantity within an order.
Reorder level	The stock threshold at or below which an item is flagged for re-purchase.
MVC	Model-View-Controller architectural pattern.
3NF	Third Normal Form (relational normalisation target).
RBAC	Role-Based Access Control.

1.5 Document Conventions

Each functional requirement is presented as a self-contained specification card containing the fields below. The keyword **shall** denotes a mandatory requirement.

Description	The behaviour the system must exhibit.
Priority	High = must-have for launch; Medium = important; Low = desirable.
Actors	Roles permitted to invoke the function.
Trigger / Pre-conditions	What starts the function and what must already be true.
Inputs	Data the function consumes, with field-level validation.
Processing & Business Rules	The logic the system applies.
Outputs / Post-conditions	Results produced and the resulting system state.
Exceptions	Error and alternate-flow handling.
Acceptance Criteria	Objective, testable pass conditions.
Traceability	Business objective(s) and business rule(s) satisfied.

2. System Overview

2.1 System Context

The RMS is the single source of truth for menu, pricing, orders, payments, tables, stock, suppliers, staff, and reservations. Human actors (Administrator, Manager, Cashier) interact through authenticated sessions. Two external parties are represented as data only: *Suppliers*, whose goods are recorded through purchase orders, and *Customers*, whose orders and reservations are recorded but who do not log in. Payment is **recorded** in the system but **processed outside** it (no card-gateway integration in scope).

2.2 Architecture & Phasing

The system is built on the Model-View-Controller pattern with an explicit service/data layer that is independent of the desktop UI framework, so the same business logic and MySQL database serve both phases. This is a functional constraint because it shapes where behaviour lives: all calculation, validation, and state rules specified in §3 must reside in the reusable business layer, not in screen code.

Phase	Interface	Data / Logic	Access
Phase 1	JavaFX desktop application, on-premise	Shared business layer + MySQL	Local machine / LAN, no internet required
Phase 2	Browser-based web application	Same business layer + same MySQL schema	More devices; groundwork for multi-location

2.3 Actors

Actor	Type	Description
Administrator	Primary	Full control: user accounts, roles, reference/system data, and every module and report. Typically the owner or IT-responsible person.
Manager	Primary	Operational control: menu & prices, tables, inventory & suppliers, reservations, staff records, and all reports. Cannot manage system-level user accounts unless also granted Administrator rights.
Cashier	Primary	Front-line: creates/manages orders, produces bills & receipts, records payments, views menu & table status, records reservations. Cannot change prices, manage stock, or view management reports.
Supplier	External (data)	Provides stock recorded through purchase orders; does not use the system.
Customer	External (data)	Experiences service; may have a reservation recorded. Does not use the system.

2.4 Role–Permission Matrix

Access to every function is governed by role (RBAC). A blank cell means the function is denied to that role. Permissions are cumulative: an Administrator implicitly holds all Manager and Cashier permissions.

Function area	Administrator	Manager	Cashier
Log in / log out	✓	✓	✓

Manage user accounts & roles	✓	—	—
Configure reference/system data (tax rate, etc.)	✓	—	—
Manage menu categories & items / prices	✓	✓	—
Define tables / capacity	✓	✓	—
View & update table status	✓	✓	✓
Create & finalise orders / bills / receipts	✓	✓	✓
Manage stock items, suppliers, purchase orders	✓	✓	—
Manage staff records	✓	✓	—
Create / manage reservations	✓	✓	✓
Generate & export reports	✓	✓	—

The matrix above is the authoritative interpretation of BRD §6. Every FR in §3 lists its permitted actors; those lists must remain consistent with this matrix and are enforced by FR-02 .

3. Functional Requirements Specification

This section expands each BRD functional requirement into a full specification. Requirements are grouped by module. Field-level input validation referenced here is consolidated in Appendix A; status values are enumerated in Appendix B.

3.1 Authentication & Administration (FR-01 ... FR-04)

FR-01 — User Login & Authentication

Module: Admin

Priority	High	Actors	All roles (Administrator, Manager, Cashier)
Trigger	User opens the application / an unauthenticated action is attempted.	Traces	BO-5; NFR-04; BR-01, BR-02

DESCRIPTION

The system shall require every user to authenticate with a username and password before any other function is accessible. No module, screen, or report is reachable without a valid, active session.

PRE-CONDITIONS

The user has an account that exists and is *active* (not deactivated). The application is running and can reach the database.

INPUTS

Username	Required. 3–50 chars. Case-insensitive match against stored accounts.
Password	Required. Entered masked. Compared against the stored one-way hash (never stored or logged in plain text).

PROCESSING & BUSINESS RULES

- Credentials are validated against the account store; the password is verified by comparing salted hashes (BR-02).
- On success the system creates an authenticated session bound to the user's role and records the login event (used by the staff activity report, FR-29).
- Deactivated accounts shall be refused even with correct credentials.
- After a configurable number of consecutive failures (default 5) the login is temporarily throttled to deter brute force.

OUTPUTS / POST-CONDITIONS

A valid session exists; the user lands on the role-appropriate home screen; the available menu reflects the user's permissions (FR-02).

EXCEPTIONS

- Unknown username or wrong password → generic message "Invalid username or password" (does not disclose which was wrong).
- Deactivated account → "This account is inactive. Contact an administrator."
- Database unreachable → "Unable to sign in — service unavailable"; no session is created.

ACCEPTANCE CRITERIA

- Given valid credentials for an active account, the user is authenticated and routed to their home screen.
- Given any invalid credential, access is denied with the generic message and no session is created.
- Passwords never appear in the database, logs, or screens in readable form.

FR-02 — Role-Based Access Control

Module: Admin

Priority	High	Actors	System (enforced for every authenticated user)
Trigger	Any attempt to open a screen or invoke a function.	Traces	BO-5; NFR-04; BR-03; §2.4 matrix

DESCRIPTION

The system shall restrict access to every function based on the logged-in user's role, per the Role–Permission Matrix (§2.4). Denied functions shall be hidden or disabled in the interface *and* re-checked on execution (defence in depth), so access cannot be gained by bypassing the UI.

PROCESSING & BUSINESS RULES

- Every function is tagged with the roles permitted to invoke it; the business layer verifies the caller's role before executing (BR-03).
- Permissions are cumulative — Administrator $\supseteq$ Manager permissions where stated; Manager and Cashier hold only their listed permissions.
- Cashiers shall not access price changes, stock management, or management reports.

OUTPUTS / POST-CONDITIONS

Permitted actions proceed; unauthorised actions are blocked and logged.

EXCEPTIONS

Unauthorised invocation → action refused with "You do not have permission to perform this action"; the attempt is recorded.

ACCEPTANCE CRITERIA

- A Cashier cannot reach price-edit, stock, or report functions by any path in the UI.
- A Manager can reach all operational functions but not user-account management (unless also an Administrator).
- Server/business-layer checks reject a role-forbidden call even if the UI control were bypassed.

FR-03 — Manage User Accounts & Roles

Module: Admin

Priority	High	Actors	Administrator
Trigger	Administrator opens User Management.	Traces	BO-5; NFR-04; BR-04, BR-05

DESCRIPTION

An Administrator shall be able to create, edit, deactivate, and delete user accounts and assign each account exactly one role (Administrator, Manager, or Cashier).

INPUTS

Username	Required, unique, 3–50 chars.
Full name	Required, 2–100 chars.
Role	Required; one of Administrator / Manager / Cashier.
Initial password	Required on create; meets password policy (Appendix A); stored hashed.
Status	Active / Inactive (default Active).

PROCESSING & BUSINESS RULES

- Usernames are unique across the system (BR-04); duplicates are rejected.
- **Deactivate** preserves the account and its historical links (orders, logins) but blocks login. **Delete** is permitted only when the account has no dependent historical records; otherwise the system requires deactivation instead (BR-05, referential integrity).
- The system shall prevent removing or deactivating the last remaining active Administrator (BR-06).
- Role changes take effect on the user's next login.

OUTPUTS / POST-CONDITIONS

The account list reflects the change; affected user's permissions update per FR-02.

EXCEPTIONS

- Duplicate username → "Username already exists."
- Delete blocked by history → offer to deactivate instead.
- Attempt to remove last Administrator → refused with explanation.

ACCEPTANCE CRITERIA

- A new account with a valid unique username and role can log in immediately.
- A deactivated account cannot log in but its past records remain intact.
- The system never allows zero active Administrators.

FR-04 — Logout & Session Termination

Module: Admin

Priority	High	Actors	All roles
Trigger	User selects Log Out (or session times out).	Traces	NFR-04; BR-07

DESCRIPTION

The system shall allow a user to log out and shall end the session on logout, returning to the login screen and clearing in-memory user context.

PROCESSING & BUSINESS RULES

- On logout the session is invalidated and the logout event is recorded (feeds FR-29).
- If an order is open and unsaved at logout, the system warns and requires the user to finalise, park, or discard it first (BR-07) so no partial financial data is orphaned (NFR-03).
- An optional idle timeout (default 15 min) triggers automatic logout to protect an unattended terminal.

OUTPUTS / POST-CONDITIONS

No authenticated session remains; the login screen is displayed; protected functions are again inaccessible.

ACCEPTANCE CRITERIA

- After logout, pressing Back or re-invoking a protected function requires re-authentication.
- Logout with an open order prompts for resolution before ending the session.

3.2 Menu & Table Management (FR-05 ... FR-09)

FR-05 — Manage Menu Categories

Module: Menu

Priority	High	Actors	Manager, Administrator
Trigger	Manager opens Menu → Categories.	Traces	BO-1, BO-5; BR-08

DESCRIPTION

A Manager shall be able to create, edit, and delete menu categories (e.g., Starters, Mains, Beverages) used to organise menu items.

INPUTS

Category name	Required, unique, 2–50 chars.
Display order	Optional integer controlling on-screen ordering.

PROCESSING & BUSINESS RULES

- Category names are unique (BR-08).
- A category that still contains menu items cannot be hard-deleted; the system requires the items to be reassigned or removed first (referential integrity).
- Changes appear immediately in the ordering screens (menu-maintenance flow).

OUTPUTS / POST-CONDITIONS

Updated category list; ordering screens reflect the change.

EXCEPTIONS

- Duplicate name → rejected. Delete of non-empty category → blocked with guidance.

ACCEPTANCE CRITERIA

- A newly created category is immediately selectable when adding a menu item and when ordering.
- Deleting a non-empty category is prevented.

FR-06 — Manage Menu Items

Module: Menu

Priority	High	Actors	Manager, Administrator
Trigger	Manager opens Menu → Items.	Traces	BO-1, BO-2, BO-5; BR-09, BR-10

DESCRIPTION

A Manager shall be able to create, edit, and delete menu items, each with a name, category, price, and availability status.

INPUTS

Item name	Required, 2–80 chars, unique within its category.
Category	Required; existing category (FR-05).
Price	Required; decimal ≥ 0.00 , two decimal places, within currency limits.
Availability	Available / Unavailable (default Available).
Description (opt.)	Up to 255 chars.

PROCESSING & BUSINESS RULES

- Price must be non-negative and is stored to two decimals; price changes apply only to *future* orders — bills already finalised are never retro-priced (BR-09).
- An item referenced by any historical order shall not be hard-deleted; it is instead marked Unavailable/ discontinued to preserve order history (BR-10; see FR-07).
- Menu changes are reflected immediately in the ordering screens.

OUTPUTS / POST-CONDITIONS

Item catalogue updated; ordering screens show current items and prices.

EXCEPTIONS

- Negative/invalid price → rejected. Delete of item with order history → converted to "mark unavailable".

ACCEPTANCE CRITERIA

- A new available item with a valid price is orderable immediately at that price.
- Editing a price does not alter any previously finalised bill.

FR-07 — Toggle Menu Item Availability

Module: Menu

Priority	Medium	Actors	Manager, Administrator
Trigger	Item runs out / returns (e.g., "86" an item).	Traces	BO-1; BR-10

DESCRIPTION

The system shall allow a menu item to be marked available or unavailable without deleting it, so seasonal or sold-out items can be hidden from ordering yet retained for history and later reactivation.

PROCESSING & BUSINESS RULES

- Unavailable items are not selectable when building a new order but remain visible (greyed/flagged) in menu administration.
- Marking unavailable does not affect items already on open orders.

OUTPUTS / POST-CONDITIONS

Ordering screens include only available items; the item can be re-enabled at any time.

ACCEPTANCE CRITERIA

- An item toggled Unavailable disappears from the order-entry list but its record and history persist.
- Re-enabling makes it immediately orderable again.

FR-08 — Define Dining Tables

Module: Tables

Priority	High	Actors	Manager, Administrator
Trigger	Manager configures the dining room.	Traces	BO-1, BO-5; BR-11

DESCRIPTION

A Manager shall be able to define dining tables, each with a unique number/label and a seating capacity.

INPUTS

Table number/label	Required, unique, 1–10 chars.
Seating capacity	Required, integer ≥ 1 .

PROCESSING & BUSINESS RULES

- Table labels are unique (BR-11); capacity supports reservation party-size checks (FR-24).
- A table currently occupied or holding future reservations cannot be deleted until it is free and un-booked.

OUTPUTS / POST-CONDITIONS

The floor/table list reflects the definition; new tables default to status **Free**.

ACCEPTANCE CRITERIA

- A defined table appears for order and reservation selection with its capacity.
- Duplicate labels are rejected.

FR-09 — Display & Update Table Status

Module: Tables

Priority	High	Actors	All roles (view/update per matrix)
Trigger	Order/reservation events, or manual status change.	Traces	BO-1; BR-12; Appendix B

DESCRIPTION

The system shall display and update each table's status. Valid statuses are **Free** , **Occupied** , **Reserved** , and **Needs Cleaning** .

PROCESSING & BUSINESS RULES (STATE MODEL — SEE APPENDIX B)

- Opening a dine-in order on a Free/Reserved table sets it **Occupied** .
- Closing a dine-in order sets the table **Needs Cleaning** (FR-17); staff mark it **Free** when cleaned.
- A confirmed reservation for the current service sets/keeps the table **Reserved** until seated or cancelled (FR-25).
- Only defined transitions are allowed; illegal transitions are rejected (BR-12).

OUTPUTS / POST-CONDITIONS

A live floor view shows current status per table, colour-coded, refreshed as events occur.

ACCEPTANCE CRITERIA

- Table status visibly changes in response to order open/close and reservation events.
- An invalid manual transition is prevented.

3.3 Orders & Billing / POS (FR-10 ... FR-17)

Financial-integrity note: All monetary calculation for FR-12 to FR-16 shall be centralised in the business layer (one billing engine), computed in a single database transaction, and rounded consistently (BR-13, BR-16). This directly mitigates the BRD risk "Incorrect financial calculations" and satisfies BO-2.

FR-10 — Open a New Order

Module: POS

Priority	High	Actors	Cashier, Manager, Administrator
Trigger	A customer is seated or places a takeaway order.	Traces	BO-1; BR-14

DESCRIPTION

A Cashier shall be able to open a new order for a specific dine-in table *or* as takeaway.

INPUTS

Order type	Required; Dine-in or Takeaway .
Table	Required if Dine-in; must be a Free or Reserved (own) table. Omitted for Takeaway.

PROCESSING & BUSINESS RULES

- A new order is created with a unique order number, status **Open** , timestamp, and the creating user (BR-14).
- For Dine-in, the selected table cannot already have an open order; on open, the table becomes **Occupied** (FR-09).

OUTPUTS / POST-CONDITIONS

An open, empty order exists and is ready to receive items; the table (if any) is Occupied.

EXCEPTIONS

- Table already has an open order → prevented, with option to open that existing order.

ACCEPTANCE CRITERIA

- A dine-in order can only be opened on a table without an active order, and that table becomes Occupied.
- A takeaway order opens with no table.

FR-11 — Add / Remove Order Items

Module: POS

Priority	High	Actors	Cashier, Manager, Administrator
Trigger	Order is open; customer selects/changes items.	Traces	BO-1, BO-2; BR-15

DESCRIPTION

A Cashier shall be able to add available menu items with quantities to an open order and remove them before finalising.

INPUTS

Menu item	Required; must be currently <i>Available</i> (FR-07).
Quantity	Required; integer ≥ 1 .

PROCESSING & BUSINESS RULES

- Each order line captures item, quantity, and the unit price *at the time of ordering* (price snapshot — BR-15), protecting the bill from later menu price edits.
- Adding an item already on the order increments its quantity (or adds a new line, per configuration).
- Items may be added or removed only while the order status is *Open*; a finalised order is immutable.
- The order subtotal recomputes automatically on every change (FR-12).

OUTPUTS / POST-CONDITIONS

The order reflects current lines and quantities with a live subtotal.

EXCEPTIONS

- Quantity < 1 or non-integer $\rightarrow$ rejected. Adding an unavailable item $\rightarrow$ blocked. Editing a finalised order $\rightarrow$ blocked.

ACCEPTANCE CRITERIA

- Adding/removing lines updates the subtotal immediately and correctly.
- Only available items can be added; finalised orders cannot be edited.

FR-12 — Automatic Subtotal Calculation

Module: POS

Priority	High	Actors	System
Trigger	Any change to order lines/quantities.	Traces	BO-2; BR-13

DESCRIPTION

The system shall calculate the order subtotal automatically from item prices and quantities — no manual arithmetic.

PROCESSING & BUSINESS RULES

- $\text{line total} = \text{unit price (snapshot)} \times \text{quantity}$; $\text{subtotal} = \sum \text{line totals}$ (BR-13).
- Rounding to two decimals is applied consistently (BR-16); computation happens in the business layer, not the UI.

OUTPUTS / POST-CONDITIONS

A current subtotal displayed on the order and carried into the bill.

ACCEPTANCE CRITERIA

- For any set of lines, the displayed subtotal equals $\sum(\text{price} \times \text{qty})$ to two decimals.
- The subtotal updates within the performance target (NFR-02) on each edit.

FR-13 — Apply Discount

Module: POS

Priority	Medium	Actors	Cashier, Manager, Administrator
Trigger	A discount is agreed before finalising.	Traces	BO-2; BR-17

DESCRIPTION

The system shall support applying a discount to an order, as either a percentage of the subtotal or a fixed amount.

INPUTS

Discount type	Percentage or Fixed amount.
Discount value	Percentage 0–100; or fixed amount $0 \leq \text{value} \leq \text{subtotal}$.

PROCESSING & BUSINESS RULES

- Discount is applied to the subtotal *before* tax: `discountable = subtotal - discount` (BR-17). Tax (FR-14) is computed on the discounted amount.
- A discount can never make the discounted amount negative; percentage capped at 100, fixed capped at subtotal.
- Optional policy: discounts above a threshold may require Manager authorisation (configurable) — recorded for audit.

OUTPUTS / POST-CONDITIONS

The bill shows subtotal, discount, and reduced pre-tax base.

EXCEPTIONS

- Percentage > 100 or fixed > subtotal → rejected with correction prompt.

ACCEPTANCE CRITERIA

- A 10% discount on a 100.00 subtotal yields a 90.00 pre-tax base.
- No discount can produce a negative total.

FR-14 — Automatic Tax & Final Total

Module: POS

Priority	High	Actors	System
Trigger	Bill is requested / order is being finalised.	Traces	BO-2; BR-18, BR-16

DESCRIPTION

The system shall calculate tax and the final total automatically using the configured tax rate (reference data set by Administrator).

PROCESSING & BUSINESS RULES

- $\text{tax} = \text{round}((\text{subtotal} - \text{discount}) \times \text{tax_rate})$; $\text{total} = (\text{subtotal} - \text{discount}) + \text{tax}$ (BR-18).
- The tax rate is a single configurable reference value; the rate in force at finalisation is stored on the order for audit and accurate reprinting (BR-18).
- Rounding to two decimals is applied once, consistently (BR-16).

OUTPUTS / POST-CONDITIONS

The bill displays subtotal, discount, tax, and grand total.

ACCEPTANCE CRITERIA

- With a 10% tax rate, a 90.00 pre-tax base yields tax 9.00 and total 99.00.
- Changing the reference tax rate does not alter any previously finalised bill.

FR-15 — Record Payment & Finalise Order

Module: POS

Priority	High	Actors	Cashier, Manager, Administrator
Trigger	Customer pays.	Traces	BO-1, BO-2; BR-19

DESCRIPTION

A Cashier shall be able to record the payment method and finalise (close) the order. Payment is *recorded*, not processed (no gateway in scope).

INPUTS

Payment method	Required; e.g., Cash, Card, Other (reference list).
Amount tendered (opt.)	For Cash; system computes change = tendered – total (≥ 0).

PROCESSING & BUSINESS RULES

- Finalisation records the payment, sets order status `Paid/Closed`, and stores the final figures (subtotal, discount, tax rate, tax, total) in one atomic transaction (BR-19, NFR-03).
- An order with zero items cannot be finalised.
- Once closed, the order becomes immutable (no further item/discount edits).

OUTPUTS / POST-CONDITIONS

Order is closed and payment recorded; a receipt is available (FR-16); a dine-in table is released (FR-17).

EXCEPTIONS

- No payment method → cannot finalise. Cash tendered < total → rejected. Transaction failure → order remains Open, no partial write (NFR-03).

ACCEPTANCE CRITERIA

- Finalising with a valid method closes the order and locks its figures.
- A failed finalisation leaves the order fully Open with no orphaned payment.

FR-16 — Generate Receipt

Module: POS

Priority	High	Actors	Cashier, Manager, Administrator
Trigger	Order finalised (FR-15).	Traces	BO-1, BO-2; BR-20

DESCRIPTION

The system shall generate a printable/exportable receipt for a finalised order.

PROCESSING & BUSINESS RULES (RECEIPT CONTENT — BR-20)

- The receipt shall include: restaurant name/header, receipt/order number, date-time, table or "Takeaway", itemised lines (name, qty, unit price, line total), subtotal, discount, tax rate & tax, grand total, payment method (and change if cash), and the serving cashier.
- The receipt reproduces the stored finalised figures exactly (no recomputation from current prices/tax).
- A finalised receipt may be re-printed/re-exported without altering data.

OUTPUTS / POST-CONDITIONS

A receipt is displayed and can be printed or exported (e.g., PDF).

ACCEPTANCE CRITERIA

- The receipt totals match the finalised bill exactly.
- Re-printing later yields an identical receipt regardless of later menu/tax changes.

FR-17 — Release Table on Order Close

Module: POS/Tables

Priority	High	Actors	System
Trigger	A dine-in order is closed (FR-15).	Traces	BO-1; BR-12

DESCRIPTION

The system shall release the table when a dine-in order is closed, setting it to **Needs Cleaning** (and then **Free** once staff mark it cleaned).

PROCESSING & BUSINESS RULES

- On close of a dine-in order the associated table transitions **Occupied** → **Needs Cleaning** automatically (BR-12).
- Takeaway orders have no table and trigger no table change.

OUTPUTS / POST-CONDITIONS

The table is freed for turnover; the floor view updates immediately.

ACCEPTANCE CRITERIA

- Closing a dine-in order changes its table to **Needs Cleaning** without manual action.
- The table becomes available for a new order once marked **Free**.

3.4 Inventory & Suppliers (FR-18 ... FR-22)

FR-18 — Manage Stock Items

Module: Inventory

Priority	High	Actors	Manager, Administrator
Trigger	Manager sets up or maintains inventory.	Traces	BO-4, BO-5; BR-21

DESCRIPTION

A Manager shall be able to create, edit, and delete stock items, each with a unit of measure and a reorder level.

INPUTS

Stock item name	Required, unique, 2–80 chars.
Unit of measure	Required; e.g., kg, L, unit, bottle.
Reorder level	Required; number ≥ 0 in the item's unit.
Quantity on hand	System-maintained; opening value on create ≥ 0 .

PROCESSING & BUSINESS RULES

- Stock item names are unique (BR-21); quantity on hand is never edited directly to fabricate stock — it changes only via recorded deliveries (FR-21) or explicit adjustments that are logged.
- A stock item referenced by any purchase-order history cannot be hard-deleted; it is deactivated to preserve history.

OUTPUTS / POST-CONDITIONS

The inventory catalogue reflects the change; reorder flags (FR-22) recompute.

EXCEPTIONS

- Duplicate name / negative reorder level → rejected. Delete with PO history → converted to deactivate.

ACCEPTANCE CRITERIA

- A new stock item appears in inventory with its unit and reorder level and participates in low-stock flagging.

FR-19 — Register & Maintain Suppliers

Module: Suppliers

Priority	Medium	Actors	Manager, Administrator
Trigger	A new supplier relationship is established.	Traces	BO-4, BO-5; BR-22

DESCRIPTION

A Manager shall be able to register and maintain suppliers used on purchase orders.

INPUTS

Supplier name	Required, unique, 2–100 chars.
Contact person / phone / email	Optional; email & phone format-validated when present.
Address (opt.)	Up to 255 chars.
Status	Active / Inactive.

PROCESSING & BUSINESS RULES

- Supplier names are unique (BR-22). A supplier referenced by purchase orders cannot be deleted; it is set Inactive instead.

OUTPUTS / POST-CONDITIONS

Supplier is selectable when creating purchase orders (FR-20).

ACCEPTANCE CRITERIA

- A registered active supplier can be chosen on a new purchase order.
- Deleting a supplier with PO history is prevented.

FR-20 — Create Purchase Order

Module: Purchasing

Priority	High	Actors	Manager, Administrator
Trigger	Stock is low / restock is needed.	Traces	BO-4; BR-23

DESCRIPTION

A Manager shall be able to create a purchase order (PO) for a supplier, listing stock items and quantities to be purchased.

INPUTS

Supplier	Required; an active supplier (FR-19).
PO lines	≥ 1 line; each = stock item (FR-18) + quantity > 0 (+ optional unit cost).
Expected date (opt.)	Valid date, today or later.

PROCESSING & BUSINESS RULES

- A PO is created with a unique number, status `Ordered`, creating user, and timestamp (BR-23).
- Creating a PO does *not* change stock on hand; stock increases only on receipt (FR-21).
- A PO must contain at least one valid line.

OUTPUTS / POST-CONDITIONS

An open PO awaiting delivery; visible in the purchasing list.

EXCEPTIONS

- No lines / quantity $\leq 0 \rightarrow$ rejected.

ACCEPTANCE CRITERIA

- A valid PO is saved with status `Ordered` and does not alter stock on hand.

FR-21 — Receive Delivery & Increase Stock

Module: Purchasing

Priority	High	Actors	Manager, Administrator
Trigger	Goods arrive against a PO.	Traces	BO-4; BR-24; NFR-03

DESCRIPTION

The system shall increase stock on hand when a purchase order's delivery is recorded as received.

INPUTS

PO reference	Required; an Ordered/partially-received PO.
Received quantities	Per line; $0 \leq \text{received} \leq \text{ordered}$ (partial receipts allowed).

PROCESSING & BUSINESS RULES

- On receipt, each stock item's quantity on hand increases by the received quantity in a single transaction (BR-24, NFR-03).
- PO status updates to **Received** (all lines fulfilled) or **Partially Received**.
- Recording receipt recomputes reorder flags (FR-22).

OUTPUTS / POST-CONDITIONS

Stock on hand reflects the delivery; PO status updated; low-stock flags refreshed.

EXCEPTIONS

- Received > ordered → rejected. Transaction failure → no stock change (no partial write).

ACCEPTANCE CRITERIA

- Receiving 10 units of an item raises its on-hand by exactly 10.
- A failed receipt leaves stock and PO status unchanged.

FR-22 — Flag Low Stock

Module: Inventory

Priority	High	Actors	System (surfaced to Manager/Admin)
Trigger	Stock quantity changes or inventory is viewed.	Traces	BO-4; BR-25

DESCRIPTION

The system shall flag stock items whose quantity on hand is at or below their reorder level, so shortages are caught early.

PROCESSING & BUSINESS RULES

- An item is flagged when **quantity on hand $\leq$ reorder level** (BR-25).
- Flagged items are visually highlighted in inventory and listed in the inventory report (FR-28).

OUTPUTS / POST-CONDITIONS

A visible low-stock indicator and a filterable list of items needing reorder.

ACCEPTANCE CRITERIA

- An item at or below its reorder level is flagged; receiving stock above the level clears the flag.

3.5 Staff & Reservations (FR-23 ... FR-26)

FR-23 — Manage Staff Records

Module: Staff

Priority	Medium	Actors	Manager, Administrator
Trigger	Hiring, role change, or offboarding.	Traces	BO-3, BO-5; BR-26

DESCRIPTION

A Manager shall be able to create, edit, and deactivate staff records (name, role, contact, status).

INPUTS

Name	Required, 2–100 chars.
Role / position	Required (e.g., Cashier, Waiter, Chef).
Contact	Phone/email, format-validated.
Status	Active / Inactive.

PROCESSING & BUSINESS RULES

- Staff records are distinct from system *user accounts* (FR-03); a staff record may or may not have a login. Deactivation retains history (BR-26).
- A staff record linked to historical activity is deactivated, not deleted, to preserve reporting integrity.

OUTPUTS / POST-CONDITIONS

The staff roster reflects the change; inactive staff are excluded from active listings.

ACCEPTANCE CRITERIA

- A new staff record is stored and appears in the roster; deactivation hides them from active lists but keeps history.

FR-24 — Create Reservation

Module: Reservations

Priority	High	Actors	Cashier, Manager, Administrator
Trigger	A customer books a table.	Traces	BO-1, BO-5; BR-27, BR-28

DESCRIPTION

A Cashier or Manager shall be able to create a reservation with customer name, contact, date, time, party size, and table.

INPUTS

Customer name	Required, 2–100 chars.
Contact	Required; phone/email, format-validated.
Date & time	Required; must be in the future (today or later, valid service time).
Party size	Required; integer ≥ 1 .
Table	Required; an existing table (FR-08).

PROCESSING & BUSINESS RULES

- Party size should not exceed table capacity; if it does, the system warns and may require override or a larger table (BR-28).
- The chosen table/time must not overlap an existing reservation (enforced by FR-26/BR-27).
- A confirmed reservation sets the table status to **Reserved** for that service window (FR-09).
- New reservations are created with status **Booked**.

OUTPUTS / POST-CONDITIONS

A stored reservation; the table reflects the booking; the reservation appears in the day's list.

EXCEPTIONS

- Past date/time → rejected. Overlap → rejected (FR-26). Party > capacity → warning/override.

ACCEPTANCE CRITERIA

- A valid future reservation on a free time-slot is saved and marks the table **Reserved**.

FR-25 — Reservation Lifecycle (Seat / Complete / Cancel)

Module: Reservations

Priority	Medium	Actors	Cashier, Manager, Administrator
Trigger	Guest arrives, dines, or cancels.	Traces	BO-1; BR-29; Appendix B

DESCRIPTION

The system shall allow a reservation to be seated, completed, or cancelled.

PROCESSING & BUSINESS RULES (STATE MODEL)

- Booked → Seated (guest arrives; table becomes Occupied, an order may be opened), Seated → Completed (visit ends), or Booked/Seated → Cancelled. An optional No-Show state may close out un-arrived bookings.
- Cancelling or completing a reservation frees the table's reserved hold, per the table state model (BR-29).
- Only defined transitions are permitted.

OUTPUTS / POST-CONDITIONS

Reservation and table statuses reflect the lifecycle stage.

ACCEPTANCE CRITERIA

- Seating a reservation frees the reserved hold and marks the table Occupied; cancelling frees the slot for rebooking.

FR-26 — Prevent Double-Booking

Module: Reservations

Priority	High	Actors	System
Trigger	A reservation is created or rescheduled.	Traces	BO-1; BR-27

DESCRIPTION

The system shall prevent double-booking the same table for overlapping reservation times.

PROCESSING & BUSINESS RULES

- Before saving, the system checks the target table for any active (Booked/Seated) reservation whose time window overlaps the requested window (default slot duration configurable) — BR-27.
- Overlaps are rejected; the system may suggest alternative free tables/times.
- Cancelled/completed reservations do not block new bookings.

OUTPUTS / POST-CONDITIONS

Only non-conflicting reservations are stored.

EXCEPTIONS

- Overlap detected → save blocked with a clear conflict message.

ACCEPTANCE CRITERIA

- Two overlapping reservations on the same table cannot both exist.
- Back-to-back non-overlapping bookings are allowed.

3.6 Reporting (FR-27 ... FR-30)

FR-27 — Sales Report

Module: Reporting

Priority	High	Actors	Manager, Administrator
Trigger	Manager selects Reports → Sales.	Traces	BO-3; BR-30

DESCRIPTION

A Manager shall be able to generate a sales report for a chosen date range showing totals and item breakdowns.

INPUTS

Date range	Required; start ≤ end; both valid dates.
Filters (opt.)	Order type, payment method, cashier, category.

PROCESSING & BUSINESS RULES

- Aggregates only *finalised* orders within the range: total sales, order count, average order value, tax collected, discounts given, and a per-item / per-category quantity-and-revenue breakdown (BR-30).
- Figures derive from stored finalised order data (never from current menu prices).

OUTPUTS / POST-CONDITIONS

An on-screen sales report, exportable via FR-30.

EXCEPTIONS

- start > end → rejected. No data in range → empty report with a clear note.

ACCEPTANCE CRITERIA

- Totals reconcile to the sum of finalised orders in the range.
- Item breakdown quantities and revenue match the underlying orders.

FR-28 — Inventory / Stock Report

Module: Reporting

Priority	High	Actors	Manager, Administrator
Trigger	Manager selects Reports → Inventory.	Traces	BO-4; BR-25

DESCRIPTION

A Manager shall be able to generate an inventory report showing current stock levels and items below reorder level.

PROCESSING & BUSINESS RULES

- Lists every active stock item with unit, quantity on hand, reorder level, and a low-stock indicator (quantity ≤ reorder level — BR-25).
- Supports filtering to "below reorder level only" to drive purchasing.

OUTPUTS / POST-CONDITIONS

An on-screen inventory report, exportable via FR-30.

ACCEPTANCE CRITERIA

- Every item at/below its reorder level is listed as low.
- Quantities match live stock on hand at generation time.

FR-29 — Staff Activity Report

Module: Reporting

Priority	Medium	Actors	Manager, Administrator
Trigger	Manager selects Reports → Staff Activity.	Traces	BO-3; FR-01, FR-04

DESCRIPTION

A Manager shall be able to generate a staff activity report for a chosen date range.

PROCESSING & BUSINESS RULES

- Summarises, per user/cashier: login/logout activity, number of orders processed, total sales handled, and discounts applied — derived from session events (FR-01/FR-04) and finalised orders.
- Respects RBAC: only Manager/Administrator may view it (FR-02).

OUTPUTS / POST-CONDITIONS

An on-screen staff activity report, exportable via FR-30.

ACCEPTANCE CRITERIA

- Order counts and sales totals per cashier reconcile to finalised orders in the range.

FR-30 — Export Reports

Module: Reporting

Priority	Medium	Actors	Manager, Administrator
Trigger	A report is displayed.	Traces	BO-3

DESCRIPTION

The system shall allow any report (FR-27...FR-29) to be exported to a shareable, printable format (e.g., PDF) or printed directly.

PROCESSING & BUSINESS RULES

- The export reproduces the on-screen report exactly, with title, generated-by user, generation timestamp, and the applied date range/filters in the header.

OUTPUTS / POST-CONDITIONS

A file/printout suitable for records or sharing with the owner.

ACCEPTANCE CRITERIA

- The exported document matches the displayed report and records its parameters.

4. Key Use Cases & Process Flows

These end-to-end flows show how the functional requirements combine to serve the operational processes described in BRD §7. Each numbered step references the FR(s) it exercises.

UC-1 — Order-to-Payment (Dine-in)

Field	Detail
Actor	Cashier
Goal	Serve a seated party from order entry to paid receipt with an accurate bill.
Pre-conditions	Cashier is authenticated (FR-01); a Free table exists.

Main flow:

- Cashier opens a new dine-in order and selects the table → table becomes Occupied. FR-10 FR-09
- Cashier adds available menu items with quantities; subtotal updates live. FR-11 FR-12
- Customer requests the bill; cashier optionally applies a discount. FR-13
- System computes tax and grand total. FR-14
- Cashier records the payment method and finalises the order (atomic). FR-15
- System generates the receipt (print/export). FR-16
- System sets the table to Needs Cleaning → Free after cleaning. FR-17 FR-09

Alternate / exception flows: item unavailable (blocked at step 2, FR-11); discount invalid (FR-13); finalisation transaction fails → order stays Open, nothing written (FR-15, NFR-03); takeaway variant skips table steps 1 & 7.

UC-2 — Menu Maintenance

- Manager authenticates and opens Menu administration. FR-01
- Manager creates/edits categories. FR-05
- Manager creates/edits items with price & availability; changes appear immediately in ordering. FR-06
- Manager marks sold-out items Unavailable (retaining history) and re-enables later. FR-07

Rule: price edits affect only future orders, never finalised bills (BR-09).

UC-3 — Inventory & Purchasing

- Manager registers suppliers and stock items with reorder levels. FR-19 FR-18
- System flags items at/below reorder level. FR-22
- Manager raises a purchase order to a supplier. FR-20
- On delivery, manager records received quantities → stock on hand increases (atomic); flags refresh. FR-21 FR-22

UC-4 — Reservation

- Cashier/Manager creates a reservation (customer, date/time, party size, table). FR-24
- System rejects overlapping bookings on the same table. FR-26

3. Table shows Reserved for the service window. FR-09
4. Reservation is seated, completed, or cancelled. FR-25

UC-5 — Reporting

1. Manager/Administrator selects a report and date range. FR-27 FR-28 FR-29
2. System aggregates the underlying data and presents results.
3. Manager exports/prints the report. FR-30

5. Data Requirements & State Models

The functions above operate over the core entities below. This is a functional-level view; full attributes, keys, and the 3NF schema are finalised in the Database Design (REF-3). All operational data resides in one shared MySQL database (NFR-07) used by both phases.

5.1 Core Entities

Entity	Purpose / key attributes (indicative)	Key relationships
User Account	Login identity: username, password hash, role, status.	Has one Role; creates Orders, POs.
Role	Administrator / Manager / Cashier; governs permissions.	Held by User Accounts.
Staff	Employee record: name, position, contact, status.	Optionally linked to a User Account.
Menu Category	Grouping: name, display order.	Has many Menu Items.
Menu Item	Sellable item: name, price, availability.	Belongs to a Category; referenced by Order Items.
Table	Dining table: label, capacity, status.	Referenced by Orders and Reservations.
Order	Header: number, type, status, timestamps, subtotal, discount, tax rate, tax, total, payment.	Belongs to a Table (dine-in); has many Order Items; has one Payment.
Order Item	Line: item, quantity, unit-price snapshot, line total.	Belongs to an Order; references a Menu Item.
Payment	Method, amount, timestamp (recorded, not processed).	Belongs to one finalised Order.
Stock Item	Ingredient/supply: name, unit, reorder level, quantity on hand.	Referenced by PO Items.
Supplier	Vendor: name, contact, status.	Has many Purchase Orders.
Purchase Order	Header: number, supplier, status, dates.	Belongs to a Supplier; has many PO Items.
Purchase Order Item	Line: stock item, ordered qty, received qty, unit cost.	Belongs to a PO; references a Stock Item.
Reservation	Customer, contact, date/time, party size, status.	Belongs to a Table.
Reference Data	Tax rate, payment methods, and other system constants.	Consumed by billing and validation.

5.2 Key Data Integrity Rules

- Financial and stock updates occur within transactions; a failed operation leaves no partial data (NFR-03).
- Records with historical references are deactivated, not deleted, to preserve auditability (menu items, suppliers, stock items, staff, accounts).
- Order Items store a unit-price snapshot; finalised orders store the tax rate in force — so history is immutable to later configuration changes.
- The schema normalises to at least 3NF (NFR-07).

5.3 State Models

Object	States	Permitted transitions
Order	Open, Paid/Closed, (Cancelled)	Open → Paid/Closed (FR-15); Open → Cancelled (voided before payment).
Table	Free, Occupied, Reserved, Needs Cleaning	Free/Reserved → Occupied (FR-10); Occupied → Needs Cleaning (FR-17); Needs Cleaning → Free; Free ↔ Reserved (FR-24/FR-25).
Reservation	Booked, Seated, Completed, Cancelled, (No-Show)	Booked → Seated → Completed; Booked/Seated → Cancelled; Booked → No-Show.
Purchase Order	Ordered, Partially Received, Received, (Cancelled)	Ordered → Partially Received → Received (FR-21); Ordered → Cancelled.

6. Business Rules Catalogue

Business rules are cross-cutting constraints referenced throughout §3. They are specified once here and enforced in the shared business layer.

ID	Rule	Enforced by
BR-01	No function is accessible without a valid authenticated session.	FR-01, FR-02
BR-02	Passwords are stored only as salted one-way hashes; never in plain text in DB, logs, or UI.	FR-01, FR-03; NFR-04
BR-03	Every function checks the caller's role in the business layer, independent of the UI.	FR-02
BR-04	Username are unique across the system.	FR-03
BR-05	Records with historical references are deactivated, not deleted.	FR-03, FR-06, FR-18, FR-19, FR-23
BR-06	At least one active Administrator must always exist.	FR-03
BR-07	Logout with an unsaved open order requires finalise/park/discard first.	FR-04
BR-08	Menu category names are unique.	FR-05
BR-09	Price changes apply to future orders only; finalised bills are never retro-priced.	FR-06, FR-14
BR-10	Items with order history are made Unavailable rather than deleted.	FR-06, FR-07
BR-11	Table labels are unique.	FR-08
BR-12	Table status follows the defined state model; illegal transitions are rejected.	FR-09, FR-17
BR-13	line total = unit-price snapshot × quantity; subtotal = Σ line totals.	FR-12
BR-14	Each order has a unique number, creating user, and timestamp.	FR-10
BR-15	Order lines capture the unit price at time of ordering (price snapshot).	FR-11
BR-16	Monetary values use two-decimal precision with one consistent rounding step.	FR-12, FR-14
BR-17	Discount applies to subtotal before tax; result can never be negative.	FR-13
BR-18	tax = (subtotal – discount) × tax_rate; total = (subtotal – discount) + tax; tax rate stored on the order.	FR-14
BR-19	Finalisation is atomic: payment + status + stored figures commit together or not at all.	FR-15; NFR-03
BR-20	Receipts reproduce stored finalised figures; re-prints are identical.	FR-16
BR-21	Stock item names are unique; on-hand changes only via receipts/logged adjustments.	FR-18, FR-21
BR-22	Supplier names are unique.	FR-19
BR-23	A PO has a unique number and ≥ 1 valid line; creating it does not change stock.	FR-20
BR-24	Receiving a delivery increases on-hand by received qty within a transaction.	FR-21

BR-25	An item is low when quantity on hand $\leq$ reorder level.	FR-22, FR-28
BR-26	Staff records are distinct from user accounts; deactivate to retain history.	FR-23
BR-27	No two active reservations may overlap on the same table.	FR-24, FR-26
BR-28	Party size exceeding table capacity triggers a warning/override.	FR-24
BR-29	Reservation lifecycle follows the defined state model and frees table holds on complete/cancel.	FR-25
BR-30	Reports aggregate only finalised data over the selected range/filters.	FR-27, FR-29

7. Non-Functional Requirements

ID	Category	Requirement & measurable target
NFR-01	Usability	Front-line ordering/billing screens shall be learnable by a cashier in a single short training session, with minimal clicks for common actions (open order, add item, bill). Target: a trained cashier completes an order-to-receipt cycle without reference to a manual.
NFR-02	Performance	Common actions (open an order, add an item, generate a bill) shall respond within ≈ 2 seconds under normal single-restaurant load.
NFR-03	Reliability & Data Integrity	Database constraints and transactions keep financial and stock data consistent; a failed operation shall leave no partial data.
NFR-04	Security	Passwords stored hashed (not plain text); access enforced by role; financial and personal data not exposed to unauthorised roles.
NFR-05	Maintainability	MVC separation of data, business logic, and presentation so the interface can be replaced (desktop $\rightarrow$ web) without rewriting business logic.
NFR-06	Portability / Migration	Business logic and data access kept independent of the desktop UI framework, reusable by the future web interface on the same MySQL database.
NFR-07	Data Persistence	All operational data stored in a MySQL relational database normalised to at least 3NF, shared across both phases.
NFR-08	Availability	For the desktop phase, the system functions on the restaurant's local machine/network without requiring internet access.

8. Assumptions, Dependencies & Constraints

8.1 Assumptions

- The restaurant operates as a single location for the initial phases.
- A reliable local computer (and network, if multi-terminal) is available for the desktop phase.
- Payment is recorded in the system but processed outside it (no card gateway in scope).
- Menu, tax rate, and reference data are provided by the restaurant during setup.
- Cashier-role staff have basic computer familiarity.

8.2 Dependencies

- Reference data (tax rate, payment methods) must be configured before billing (FR-14/FR-15).
- Menu categories/items and tables must exist before orders and reservations can be taken.
- Suppliers and stock items must exist before purchase orders and receipts.

8.3 Constraints (from BRD §12)

- **Technology:** Desktop client in Java/JavaFX; MySQL data store; MVC architecture.
- **Migration:** Business logic must not be locked into the desktop UI, so a web front end can reuse it.
- **Budget/effort:** Built by a small team, favouring proven, well-documented technologies over experimental ones.

8.4 Risk Alignment

BRD Risk	How this FRD mitigates it
Scope creep from out-of-scope features	§1.2 fixes the in/out-of-scope boundary; no out-of-scope FRs are specified.
Desktop logic too coupled to UI	NFR-05/06 and §2.2 mandate calculation/validation in a reusable business layer.
Data model changes late	§5 fixes entities/state models before heavy coding; schema to 3NF.
Incorrect financial calculations	Centralised billing engine, atomic transactions, and explicit BR-13/16/17/18/19 with worked acceptance criteria.

9. Requirements Traceability Matrix

Each functional requirement traces up to at least one business objective and across to the use case that exercises it, giving full coverage from business need to verifiable behaviour.

FR	Title	Business Objective(s)	Use Case	Priority
FR-01	Login & Authentication	BO-5	All	High
FR-02	Role-Based Access Control	BO-5	All	High
FR-03	Manage User Accounts	BO-5	Admin	High
FR-04	Logout / Session End	BO-5	All	High
FR-05	Manage Menu Categories	BO-1, BO-5	UC-2	High
FR-06	Manage Menu Items	BO-1, BO-2, BO-5	UC-2	High
FR-07	Toggle Item Availability	BO-1	UC-2	Medium
FR-08	Define Tables	BO-1, BO-5	UC-1, UC-4	High
FR-09	Table Status	BO-1	UC-1, UC-4	High
FR-10	Open Order	BO-1	UC-1	High
FR-11	Add/Remove Items	BO-1, BO-2	UC-1	High
FR-12	Subtotal Calculation	BO-2	UC-1	High
FR-13	Apply Discount	BO-2	UC-1	Medium
FR-14	Tax & Total	BO-2	UC-1	High
FR-15	Record Payment / Finalise	BO-1, BO-2	UC-1	High
FR-16	Generate Receipt	BO-1, BO-2	UC-1	High
FR-17	Release Table	BO-1	UC-1	High
FR-18	Manage Stock Items	BO-4, BO-5	UC-3	High
FR-19	Manage Suppliers	BO-4, BO-5	UC-3	Medium
FR-20	Create Purchase Order	BO-4	UC-3	High
FR-21	Receive Delivery	BO-4	UC-3	High
FR-22	Flag Low Stock	BO-4	UC-3	High
FR-23	Manage Staff	BO-3, BO-5	—	Medium
FR-24	Create Reservation	BO-1, BO-5	UC-4	High
FR-25	Reservation Lifecycle	BO-1	UC-4	Medium
FR-26	Prevent Double-Booking	BO-1	UC-4	High
FR-27	Sales Report	BO-3	UC-5	High
FR-28	Inventory Report	BO-3, BO-4	UC-5	High
FR-29	Staff Activity Report	BO-3	UC-5	Medium

FR-30	Export Reports	BO-3	UC-5	Medium
-------	----------------	------	------	--------

BO-1 Speed up service · BO-2 Reduce billing errors · BO-3 Management visibility · BO-4 Control stock & cost · BO-5 Centralise data · BO-6 Prepare for growth (served by NFR-05/06 architecture, not a single FR).

10. Acceptance & Success Criteria

The FRD is satisfied when all of the following hold, consistent with BRD §14:

1. A cashier can complete an entire order-to-receipt cycle within the system (FR-10...FR-17).
2. A manager can maintain the menu, stock, suppliers, and reservations and generate accurate, exportable reports (FR-05...FR-08, FR-18...FR-30).
3. All monetary calculations (subtotal, discount, tax, total) are computed automatically and verified correct against the worked examples in FR-12/13/14.
4. Every data operation persists correctly in a 3NF MySQL database, with no partial writes on failure (NFR-03, NFR-07).
5. Access is correctly governed by role for all functions per the §2.4 matrix (FR-01, FR-02).
6. The architecture allows the Phase 2 web front end to be added without rewriting the business or data layers (NFR-05, NFR-06).

Each acceptance criterion in §3 is written to be objectively testable and maps to test cases in the Test Plan (REF-4).

Appendix A — Field Validation Reference

Field	Rule	On violation
Username	Required, unique, 3–50 chars, no leading/trailing spaces.	Reject; "Username already exists" / "Enter 3–50 characters".
Password	Meets policy: min 8 chars, at least one letter and one digit; stored hashed.	Reject with policy hint.
Names (menu/item/table/supplier/stock/customer/staff)	Required, length-bounded, unique where specified.	Reject with field-specific message.
Price / unit cost	Decimal ≥ 0.00 , two decimals, within currency max.	Reject; "Enter a valid non-negative price".
Quantity (order / PO / party size)	Integer ≥ 1 (received qty ≥ 0).	Reject; "Enter a whole number ≥ 1 ".
Discount %	0–100.	Reject; cap to valid range.
Discount fixed	$0 \leq \text{value} \leq \text{subtotal}$.	Reject; "Discount cannot exceed subtotal".
Reorder level	Number ≥ 0 in item's unit.	Reject; "Enter a non-negative value".
Reservation date/time	Valid date/time, today or future, within service hours.	Reject; "Choose a future date/time".
Email / phone	Format-validated when provided.	Reject; "Enter a valid email/phone".
Date range (reports)	start $\leq$ end; both valid dates.	Reject; "Start date must be on or before end date".

Appendix B — Status & Enumeration Reference

Enumeration	Values
User Role	Administrator, Manager, Cashier
Account / Staff / Supplier Status	Active, Inactive
Menu Item Availability	Available, Unavailable
Table Status	Free, Occupied, Reserved, Needs Cleaning
Order Type	Dine-in, Takeaway
Order Status	Open, Paid/Closed, Cancelled
Payment Method	Cash, Card, Other (reference list)
Purchase Order Status	Ordered, Partially Received, Received, Cancelled
Reservation Status	Booked, Seated, Completed, Cancelled, No-Show

— End of Functional Requirements Document (RMS-FRD v1.0) —